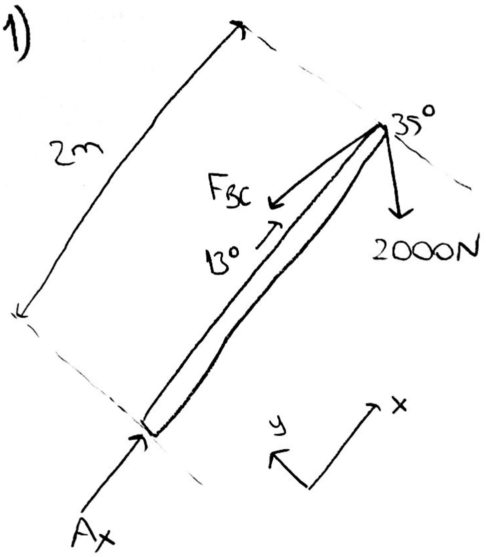
Niron CURNA
17052054

1)



Denge denklemi:

$\sum M_A = 0$

$F_{BC} \sin 13°(2) - 2000 \cos 35° (2) = 0$

$F_{BC} = 7282.94\,N$

$\sum F_x = 0;\; A_x - 7282.94 \cos 13° - 2000 \sin 35° = 0$

$A_x = 8243.43\,N$

$\sum F_y = 0;\; A_y + 7282.94 \sin 13° - 2000 \cos 35° = 0$

$A_y = 0$

$F_A = A_x = 8243.43\,N$

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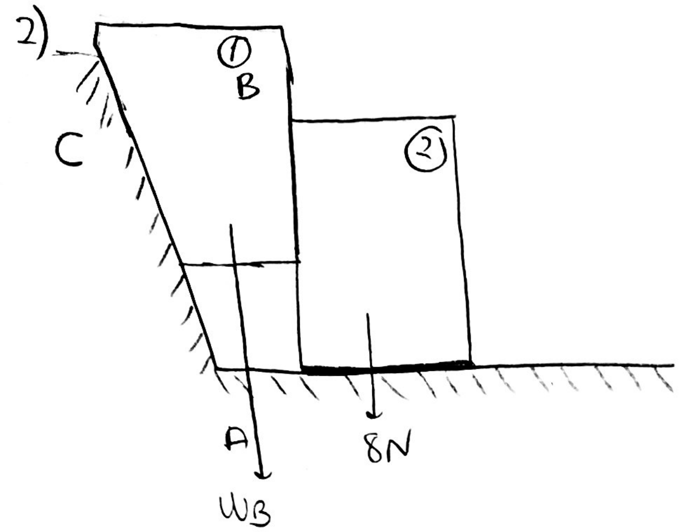
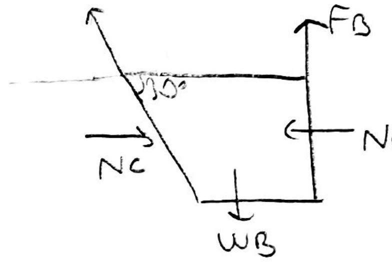
2)



$M_s = 0.5 \rightarrow A, C$

$M_s' = 0.6 \rightarrow B$

1. nin serbest cisim diyagramı



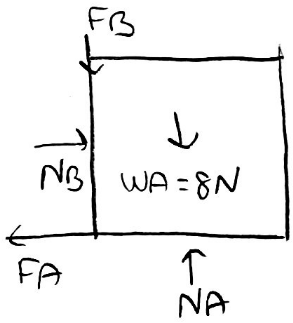
I. $\sum F_x = 0 \quad -F_C \cdot \sin 30° - N_B + N_C \cdot \cos 30° = 0$

II. $\sum F_y = 0 \quad F_C \cdot \cos 30° + F_B - W_B + N_C \cdot \sin 30° = 0$

III. $F_C = 0.5 N_C$

$F_B = 0.6 N_B$

2. nin serbest cisim diyagramı



I. $\sum F_x = 0 \quad N_B - F_A = 0$

$N_B = 0.5 N_A$

$N_A = 8 + 0.6 N_B$

$N_A = 8 + 0.6(5.714) = 11.428N$

II. $\sum F_y = 0 \quad -F_B - 8 + N_A = 0$

III. $F_B = 0.6 N_B$

$F_A = 0.5 N_A$

$N_B = 0.5(8 + 0.6 N_B)$

$N_B = 4 + 0.3 N_B$

$0.7 N_B = 4$

$N_B = 5.714N \rightarrow F_B = 3.428N$

$N_A = 11.428N \quad F_A = 5.714N$

1 ve 2 den

$-0.5 N_C \cdot \sin 30 - 5.714 + N_C \cdot \cos 30° = 0$

$0.616 N_C = 5.714$

$N_C = 9.275N \rightarrow F_C = 4.637N$

$4.637 \cdot \cos 30° + 3.428 - W_B = 0$

$W_B = 7.444N$

3)

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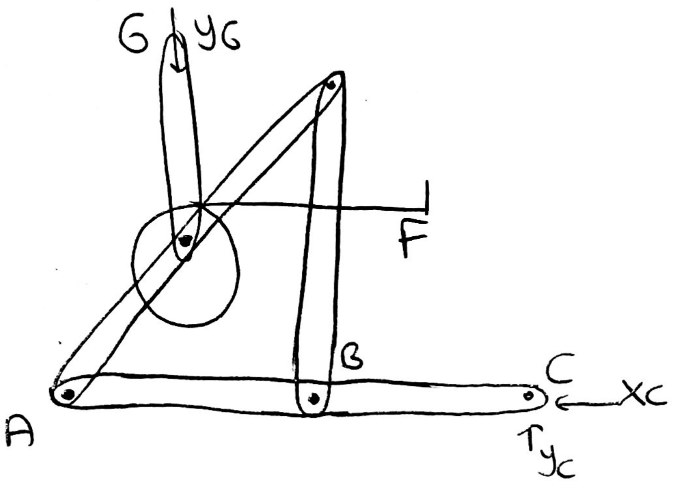
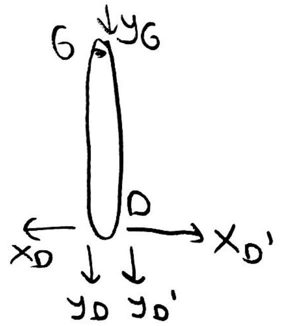


$\sum F_x = 0 \rightarrow X_C = 0$

$\sum F_y = 0 \rightarrow y_C - y_G - 100 = 0$

$\sum M_C = 0 \rightarrow 100.11 + y_G - 10 = 0$

$y_G = -\frac{1100}{10} = -110$ ve $y_C = -10$



$\sum F_x = 0$

$\sum F_y = 0$

$\sum F_y = 0$

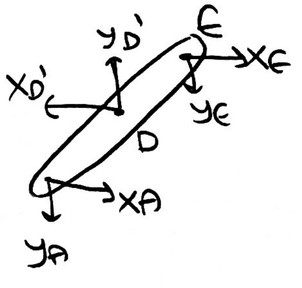
$-y_G - y_D - y_D' = 0$

$y_D' = 100 + 110$

$y_D' = 210N$

$x_D' - x_D = 0$

$x_D' = -100N$



$\sum M_E = 0$

$x_A . 6 + y_A . 8 - x_D' . 3 + y_D' . 4 = 0$

$6 . x_A - (7.5) . B + 100 . 2\ 210 . 4 = 0$

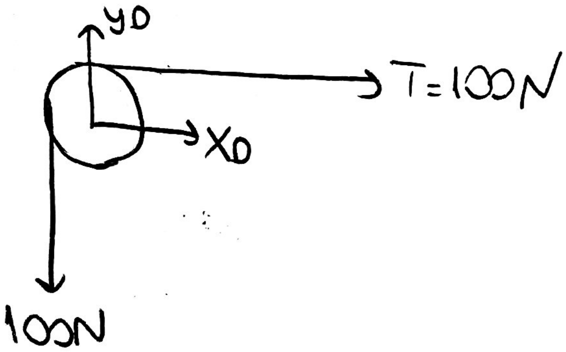
$x_A = 100N$

$\sum F_x = 0$

$x_E - x_D' + x_A = 0$

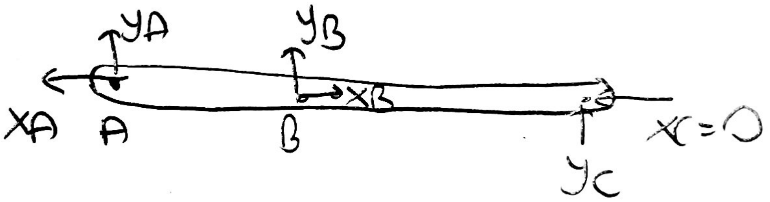
$x_E = -200N$

3) Devamı



$\Sigma F_x = 0 \quad 100 + x_D = 0 \rightarrow x_D = -100N$

$\Sigma F_y = 0 \quad y_D - 100 = 0 \rightarrow y_D = 100N$



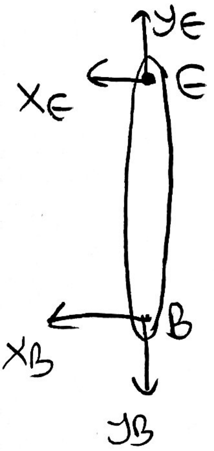
$\Sigma m_B = 0 \quad y_C\,6 - y_A\,8 = 0$

$y_A = -7{,}5N$

$\Sigma F_y = y_A + y_B + y_C = 0$

$y_B = 7{,}5 + 10$

$y_B = 17{,}5N$



$\Sigma F_x = -x_E - x_B = 0$

$\boxed{x_B = -x_E}$

$\Sigma F_y = y_E - y_B = 0$

$x_B = 200N$

$y_E = 17{,}5N$

4)

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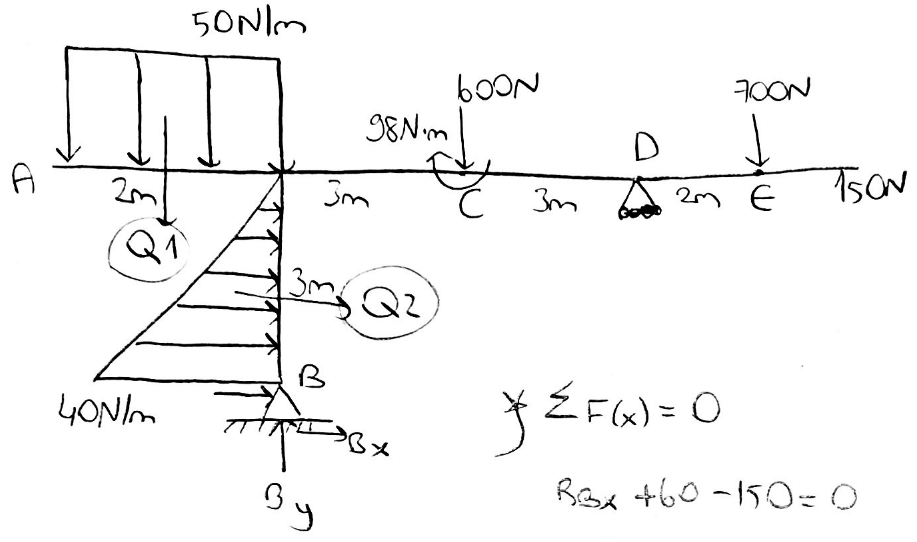

$Q_1 = 50\frac{N}{m} \cdot 2m$

$Q_1 = 100N$

$Q_2 = 40\frac{N}{m} \cdot \frac{3m}{2}$

$Q_2 = 60N$

$\sum F(x) = 0$

$R_{Bx} + 60 - 150 = 0$

$R_{Bx} = 90N$

$\sum F(y) = 0$

$R_{By} - 100 - 600 - 700 + D_y = 0$

$R_{By} + D_y - 1400 = 0$

$R_{By} + D_y = 1400N$

$R\ B_y = -292N$

$\overset{+}{\curvearrowleft} \sum M_B = 0$

$100N.1m - 60N.1m - 98N.m - 600N.3m + D_y\,6m - 700N.8m + 150$

$D_y . 6 = -7008$

$D_y = -1168N$

$R_B = \sqrt{R_{Bx}^2 + R_{By}^2}$

$R_B = \sqrt{90^2 + (-292)^2}$

$R_B = 54.6N$